

DISEASE RISK THRESHOLDS & RISK BANDS

Methodology, Rationale & Industry Validation

A Technical Reference for the Spornado Crop Disease Prediction System

1 EXECUTIVE SUMMARY

Standard machine-learning classifiers default to a 0.5 decision threshold that implicitly treats a missed outbreak and an unnecessary spray as equally costly errors. In crop disease management this assumption is economically indefensible: a missed outbreak can destroy an entire harvest while a false alarm costs one preventive spray. This document explains why the Spornado model replaces that default with a recall-constrained threshold, and how the three-zone risk band is derived. The asymmetric cost framework underpinning this model -- prioritising missed outbreak detection above all other metrics -- is not domain-specific: it is the same decision-theoretic foundation used in FDA-regulated medical diagnostics, financial fraud detection, IEC 61508 industrial safety systems, and cybersecurity intrusion detection. Within agriculture specifically, the methodology is validated by peer-reviewed systems including the FHB Risk Tool (wheatcab.psu.edu), EIPRE, and published frameworks by Hughes & Burnett (2017) and Giroux et al. (2016).

2 WHY STANDARD THRESHOLDS FAIL IN PLANT DISEASE FORECASTING

A binary classifier minimising overall error rate treats every mistake equally. In plant pathology this produces dangerous under-alerting: spray advisories are withheld to avoid false positives at the expense of missing real outbreaks.

2.1 Asymmetric Economic Costs

The cost of a false negative (missed outbreak) vastly exceeds the cost of a false positive (unnecessary spray). Elkan (2001) [1] formalises this: the optimal decision threshold is $C(FP) / [C(FP) + C(FN)]$, which collapses well below 0.5 when false negatives are more expensive.

Error Type	Agronomic Consequence	Indicative Cost
False Negative (missed outbreak)	Disease spreads unchecked; significant yield loss; potential secondary spread	USD 200-800 / acre (20-80% yield loss) [7]
False Positive (unnecessary alert)	Single preventive fungicide application; no crop damage incurred	USD 15-40 / acre (one spray) [7]

KEY INSIGHT

**Cost ratio 10:1 to 20:1 in favour of avoiding missed outbreaks.
Optimal threshold collapses well below 0.5 (Elkan, 2001 [1])**

2.2 Class Imbalance Makes ROC Misleading

The Spornado dataset has a 73% positive base rate (disease present). Saito & Rehmsmeier (2015) [3] demonstrate that under class imbalance the Precision-Recall curve is more informative than ROC because it is insensitive to the large pool of true negatives that artificially inflate ROC-AUC. Threshold selection is therefore performed on the Precision-Recall curve, not the ROC curve.

3 THEORETICAL FOUNDATION

3.1 Recall-Constrained Threshold Optimisation

The deployment threshold τ^* is defined as the solution to the constrained optimisation below, with $R_{\min} = 0.90$ (configurable). The training data is divided into three non-overlapping chronological slices: (1) a model-fit window (oldest ~80% of training) used to train and calibrate the pipeline; (2) a threshold-tuning window (newest ~20% of training) on which sklearn's `precision_recall_curve` is run to select τ^* -- this window is train-only and is never seen by the model during fitting; and (3) a fully held-out test set that is untouched until final evaluation. The test set is used exclusively to report precision, recall and AUC at the pre-chosen τ^* -- no threshold adjustment is made after the test set is observed.

OPTIMISATION FORMULA

$$\tau^* = \operatorname{argmax} \operatorname{Precision}(\tau) \quad \text{subject to: } \operatorname{Recall}(\tau) \geq R_{\min} (0.90)$$

This mirrors the sensitivity-first paradigm in medical diagnostic screening (Zweig & Campbell, 1993 [4]) and biosecurity early-warning models (Magarey et al., 2007 [5]), where missing a true positive event carries consequences that vastly outweigh the cost of a false alarm.

3.2 Temporal Splits -- Eliminating Leakage Bias

Two distinct leakage controls are applied. First, model selection uses `TimeSeriesSplit` k-fold cross-validation (strictly chronological folds) to produce unbiased AUC and F1 estimates during training -- no future observations can inform a past fold's score. Second, and separately, threshold tuning is performed on a dedicated chronological window carved from the training period (the newest ~20%), not from the final test set -- ensuring τ^* is never chosen by peeking at the data used to report it. De Wolf & Isard (2007) [6] identify temporal leakage as the most common source of optimistic bias in plant disease forecasting models; this two-stage separation directly addresses both leakage vectors they describe.

3.3 Champion Model Selection Policy

Models are ranked by a four-step hierarchy before any threshold is applied:

1. **Recall gate** Discard any model whose $\text{threshold_achieved_recall} < R_{\min}$.
2. **Highest recall** Among feasible models, prefer the one that catches more outbreaks.
3. **Highest precision** Tie-break by minimising false alerts at the operating threshold.
4. **AUC tie-break** If precision is identical, select the model with higher ROC-AUC.

4 RISK BAND ARCHITECTURE

A binary HIGH / NOT-HIGH alert provides insufficient resolution: probabilities of 0.51 and 0.99 would trigger identical instructions. The three-zone band converts a continuous probability score into graduated management guidance -- analogous to traffic-light frameworks used by EFSA in pest risk categorisation [8] and by the U.S. EPA in tiered ecological risk assessment [9].

4.1 Band Derivation

- **HIGH threshold (τ^*)** The recall-constrained optimal point derived from the Precision-Recall curve (Section 3.1). Any predicted probability $\geq \tau^*$ triggers a HIGH alert.
- **LOW threshold (τ_{low})** $= \tau^* \times \text{medium_band_factor}$ A proportional boundary set at a fixed fraction of τ^* (default 0.70). Probabilities below τ_{low} indicate safe conditions -- the model is expressing genuine low-risk confidence.
- **MEDIUM band** $= [\tau_{\text{low}}, \tau^*)$ The watch zone. Conditions are trending toward the action threshold but have not yet reached it. Growers should scout fields and prepare materials.

The $\text{medium_band_factor}$ (default 0.70) is an operational parameter, not a statistical result. It defines the width of the watch zone as a proportion of the decision boundary -- a factor of 0.70 means the MEDIUM band spans the bottom 30% of τ^* . This can be adjusted with the client to reflect operational lead-time requirements: a wider band (lower factor) provides earlier warning; a narrower band (higher factor) reduces unnecessary monitoring load. As seasonal data accumulates, the factor can be validated against empirical lead times between MEDIUM onset and confirmed outbreaks.

4.2 Risk Band Reference Table

Risk Zone	Probability Range	Recommended Action	Rationale
LOW	0.00 to τ_{low}	No action required. Conditions unfavourable for disease.	<i>Probability well below boundary; fungicide yields no expected benefit.</i>
MEDIUM	τ_{low} to τ_{high}	Scout fields. Prepare fungicide. Monitor weather closely.	<i>Model uncertain; ground scouting reduces unnecessary spray risk.</i>
HIGH	τ_{high} to 1.00	Apply registered fungicide. Alert neighbouring field managers.	<i>Exceeds recall-constrained threshold; outbreak likely imminent.</i>

5 INDUSTRY ADOPTION & CROSS-INDUSTRY VALIDATION

The asymmetric cost argument -- $C(FN) \gg C(FP)$ -- is not an agricultural concept. It is the correct decision-theoretic framing for any domain where one type of error carries irreversible consequences. Recall-constrained threshold optimisation with graduated risk bands is a mature, battle-tested pattern across multiple industries. The Spornado model applies the same rigorous methodology used in medical diagnostics, financial fraud detection, industrial safety, and cybersecurity.

Organisation	System / Product	Parallel with This Model
Bayer Digital Farming (The Climate Corporation)	Climate FieldView [11]	Uses machine learning models for disease risk prediction with multi-level severity categorisation (mild / moderate / severe) and weather-based forecasting. Exact probability scoring and threshold methodology are proprietary.
Syngenta	Cropwise Platform [12]	Provides disease monitoring and management recommendations built on agronomic models and 20+ years of historical weather data. Specific internal alert calibration methodology is not publicly documented.
USDA APHIS & NC State University	NAPFAST -- National Plant Pest Forecasting System [5]	Peer-reviewed GIS-based system using weather-driven infection-period models for plant pathogen risk assessment. Threshold exceedance triggers graduated management advisories -- published in <i>Plant Disease</i> (2007).
DTN / Progressive Farmer	DTN Agronomic Insights [13]	Field-level disease alert platform with customisable threshold-based alerting for pests and disease symptoms. Supports tiered monitoring; specific ML methodology is not publicly documented.
Penn State / USDA (1975)	BLITECAST -- Potato Late Blight Forecast System [14]	Foundational computerised forecast combining Hyre and Wallin epidemiological models; accumulated severity values trigger spray advisories. Basis for many modern late blight decision-support systems including BlightPro.

5.1 Peer-Reviewed Validated Systems

Beyond commercial platforms, the sensitivity-first threshold paradigm is well established in the peer-reviewed epidemiological literature. The four systems below have been independently validated in production environments and share the core logic of the Spornado framework: select the operating threshold by constraining sensitivity (recall) rather than optimising aggregate accuracy.

System & Institution	Key Publication	Parallel with Spornado Model
FHB Risk Assessment Tool wheatcab.psu.edu (Kansas State / Ohio State / Penn State -- USWBSI)	De Wolf, Madden & Lipps (2003). <i>Phytopathology</i> 93(4):428-435. [15]	Logistic regression model calibrated to maximise sensitivity at the operating threshold; achieves $\geq 95\%$ sensitivity in prospective validation. Champion model selected on sensitivity -- directly analogous to Spornado's recall-constrained champion-selection policy.

EIPRE Decision-Support System (Wageningen University / Dutch Ministry of Agriculture)	Zadoks (1981). EPPO Bulletin 11:365-369. [16] Reinink (1986). Neth. J. Plant Pathol. 92:3-14. [17]	Economic threshold logic: spray action triggered only when expected loss from inaction exceeds treatment cost. Conceptual ancestor of recall-constrained optimisation -- asymmetric cost structure is mathematically equivalent.
Sensitivity / Specificity Framework for Disease Warning Systems (Hughes & Burnett, 2017)	Hughes & Burnett (2017). Phytopathology 107(10):1136-1143. [18]	Formal statistical framework explicitly advocating threshold selection based on sensitivity (recall) constraints in plant disease advisory systems. Directly validates the approach used in this model.
ROC-Based Threshold Re-optimisation for FHB Warning System (Giroux et al., 2016)	Giroux et al. (2016). Plant Disease 100(6):1192-1201. [19]	Uses ROC / precision-recall curve analysis to re-select FHB warning thresholds that maximise sensitivity for a given specificity level -- the same mathematical approach as Spornado's argmax Precision s.t. Recall $\geq R_{\min}$ search.

5.2 Cross-Industry Adoption of the Same Paradigm

The table below shows that recall-constrained threshold design is not a niche agricultural technique -- it is the standard engineering response to asymmetric costs in any high-stakes classification system. Each domain independently arrived at the same mathematical structure: set a sensitivity floor first, then optimise precision within that constraint.

Domain	System / Standard	Asymmetric Cost Logic
Medical Diagnostics	FDA Diagnostic Device Guidance (2007) [20]	FDA statistical guidance for in-vitro diagnostic devices requires sensitivity and specificity to be reported and evaluated separately. For screening tests (cancer, infectious disease), sensitivity constraints are set as primary requirements -- the direct equivalent of Spornado's recall floor.
Financial Fraud Detection	Asymmetric Cost Classification [1]	False Negative (missed fraud) = customer financial loss + regulatory exposure. False Positive (blocked legitimate transaction) = customer friction. Financial institutions apply cost-sensitive threshold optimisation (Elkan, 2001) to deliberately operate below 0.5 -- identical mathematical framing to this model. Internal thresholds are proprietary.
Industrial Safety Systems	IEC 61508 Functional Safety Standard [21]	Safety Integrity Levels (SIL 1-4) define maximum tolerable Probability of Dangerous Failure per Hour (PFDh) -- a regulatory false-negative ceiling. Sensors and actuators in nuclear, chemical and rail applications must meet these sensitivity floors before any other performance metric is considered.
Cybersecurity / Intrusion Detection	Published IDS Research (Axelsson, 2000) [22]	Axelsson (2000) formally demonstrated that IDS systems must be calibrated for high sensitivity (recall) to overcome the base-rate fallacy -- low attack frequency means even small false-negative rates cause most real attacks to be missed. Security operations centres configure detection thresholds accordingly.

Drug Safety Surveillance	FDA MedWatch / EMA EudraVigilance	Post-market pharmacovigilance uses Proportional Reporting Ratio (PRR) and similar signal-detection algorithms explicitly tuned for high sensitivity over precision. A missed adverse drug event signal (FN) carries patient safety and regulatory consequences that vastly exceed the cost of a false signal (FP). [Internal algorithm parameters are published in regulatory guidance.]
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6 CURRENT MODEL PERFORMANCE AT OPERATING THRESHOLDS

All metrics are computed at the recall-constrained deployment threshold on the chronological hold-out test set -- not at the sklearn default of 0.5.

Crop	Champion Model	Threshold tau*	Recall	Precision	ROC-AUC
Corn	LogisticRegression (baseline)	0.177	94.8%	83.3%	0.834
Soybean	LogisticRegression (baseline)	0.682	95.2%	97.5%	0.911
Wheat	LogisticRegression (baseline)	0.283	100%	75.9%	0.680

Wheat's lower AUC (0.680) reflects a limited training set (~370 rows). All three crops satisfy the $\geq 90\%$ recall constraint at their respective operating thresholds. Confidence intervals will tighten as additional trap data is collected for the 2026 season.

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